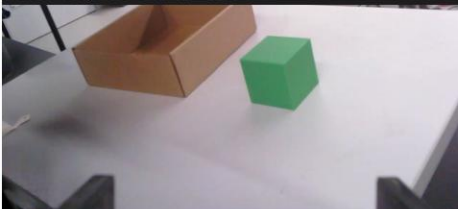
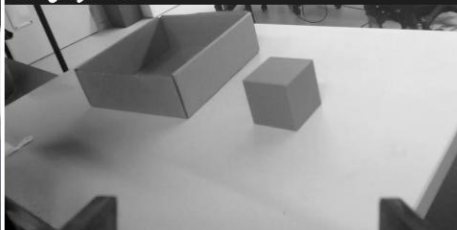
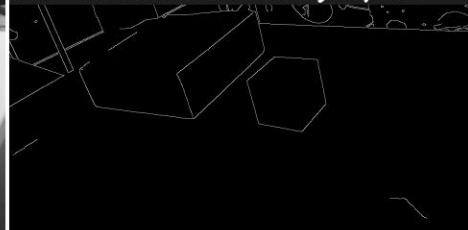
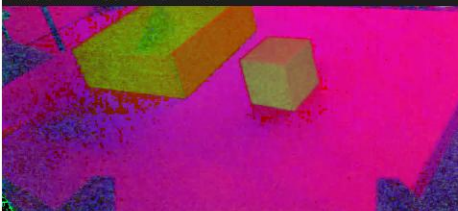


01 colour frame

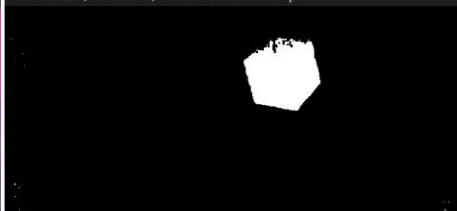
1280x720, left wrist camera

**02 greyscale****03 Gaussian blur 5x5 + Canny 60/160****04 HSV (shown as BGR)**

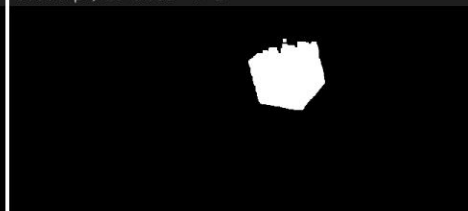
median H 133 S 16 V 163

**05 green threshold**

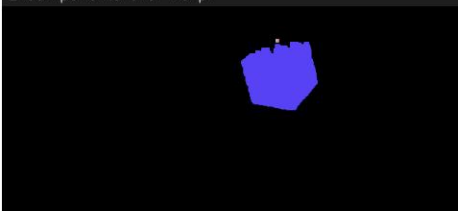
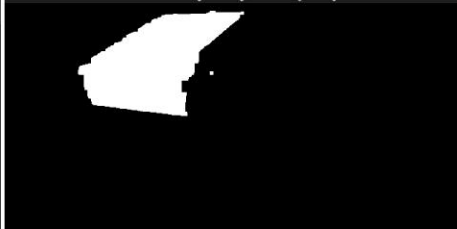
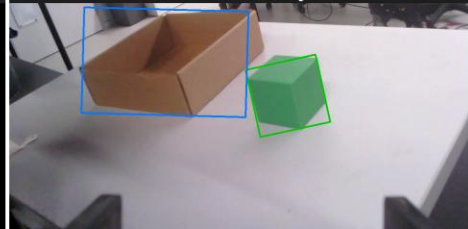
H 40-85, S>=80, V>=40: 31333 px

**06 morphological open 9x9**

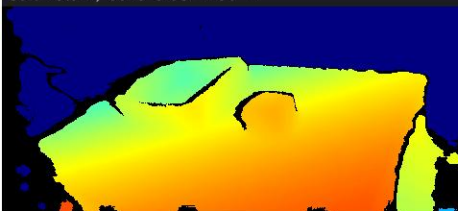
30550 px, 26 blobs -> 2

**07 connected components (8-conn)**

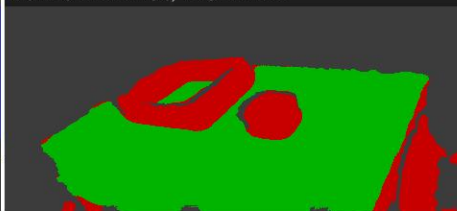
2 components over 40 px

**08 box threshold (HSV) + open/close****09 minAreaRect on each object****10 depth (Kinova module, 480x270)**

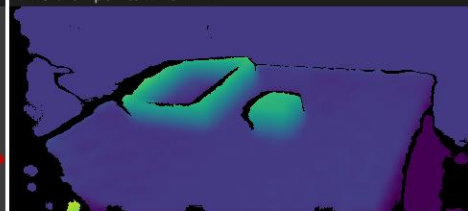
89% return; bond 0.08-1.50 m

**11 RANSAC table plane, 400 draws**

within 6 mm: 74.0%, RMS 1.4 mm

**12 height above the plane (mm)**

17.5% of points > 5 mm

**13 pixels more than 5 mm above the plane**

2 pieces > 200 px (pipeline gate: 4)

**14 result: objects and grasp check**

cube 60 mm fits 85 mm gripper; box 341 mm not

